Essentials of Data Science Laboratory - 2304102L - 2026

Search course ctrl + k
3. Practical 3
3.1. Practice Lab Assignment
3.1.1. Numpy Array Operations
3.2. Lab Assignment
3.2.1. Numpy: Matrix Operations
3.2.2. Numpy: Horizontal and Vertical Stac...
3.2.3. Numpy: Custom Sequence Generation
3.2.4. Numpy: Arithmetic and Statistical Op...
3.2.5. Numpy: Copying and Viewing Arrays
3.2.6. Numpy: Searching, Sorting, Countin...
3.2.7. Student Data Analysis and Operations
4. Practical 4
5. Practical 5

3.1.1. Numpy Array Operations

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, r and c, representing the number of rows and the number of columns.
- The next r lines contain c space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

numpyarr...

```python
import numpy as np

r, c = map(int, input().split())

# Input elements row by row
elements = []
for _ in range(r):
    row = list(map(int, input().split()))
    elements.extend(row)
```

Average time 0.032 s 31.75 ms
Maximum time 0.053 s 53.00 ms

2 out of 2 shown test case(s) p
10 out of 10 hidden test case(s

Test case 1 47 ms Debug

Expected output	Actual output
3 4	3 4
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
9 10 11 12	9 10 11 12
[[1 2 3 4]	[[1 2 3 4]
[5 6 7 8]	[5 6 7 8]

Terminal Test cases

CODETANTRA Home Learn Anywhere 202501040161@mitaoe.ac.in Support Logout

Essentials of Data Science Laboratory - 2304102L - 2026 Pin

Search course ctrl + k

- 3. Practical 3
  - 3.1. Practice Lab Assignment
    - 3.1.1. Numpy Array Operations
  - 3.2. Lab Assignment
    - 3.2.1. Numpy: Matrix Operations
    - 3.2.2. Numpy: Horizontal and Vertical Stac...
    - 3.2.3. Numpy: Custom Sequence Generation
    - 3.2.4. Numpy: Arithmetic and Statistical Op...
    - 3.2.5. Numpy: Copying and Viewing Arrays
    - 3.2.6. Numpy: Searching, Sorting, Countin...
    - 3.2.7. Student Data Analysis and Operations
- 4. Practical 4
- 5. Practical 5

3.2.1. Numpy: Matrix Operations

26:58

The given code takes two 3×3 matrices, matrix_a, and matrix_b, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition** $(\text{matrix\_a} + \text{matrix\_b})$
2. **Subtraction** $(\text{matrix\_a} - \text{matrix\_b})$
3. **Element-wise Multiplication** $(\text{matrix\_a} * \text{matrix\_b})$
4. **Matrix Multiplication** $(\text{matrix\_a} \cdot \text{matrix\_b})$
5. **Transpose of Matrix A**

Input Format:

- The user will input 3 rows for matrix_a, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for matrix_b, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.

Explorer matrixOp... Submit Debugger

```python
import numpy as np

# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int,
input().split())) for i in range(3)])

print("Enter Matrix B:")
matrix_b = np.array([list(map(int,
```

Average time	Maximum time
0.044 s	0.051 s
44.50 ms	51.00 ms

2 out of 2 shown test case(s) p

2 out of 2 hidden test case(s) p

Test case 1 51 ms Debug

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
1 1 1	1 1 1

Terminal Test cases

Prev Reset Submit Next

CODETANTRA Home Learn Anywhere 202501040161@mitaoe.ac.in Support Logout

Essentials of Data Science Laboratory - 2304102L - 2026

Pin

Search course ctrl + k

3. Practical 3
- 3.1. Practice Lab Assignment
  - 3.1.1. Numpy Array Operations
- 3.2. Lab Assignment
  - 3.2.1. Numpy: Matrix Operations
  - 3.2.2. Numpy: Horizontal and Vertical Stac...
  - 3.2.3. Numpy: Custom Sequence Generation
  - 3.2.4. Numpy: Arithmetic and Statistical Op...
  - 3.2.5. Numpy: Copying and Viewing Arrays
  - 3.2.6. Numpy: Searching, Sorting, Countin...
  - 3.2.7. Student Data Analysis and Operations

4. Practical 4

5. Practical 5

3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

07:39

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking**: Stack the two matrices horizontally (side by side).
- **Vertical Stacking**: Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases +

Explorer

stacking.py Submit

```python
import numpy as np

# Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split()))
for i in range(3)])

print("Enter Array2:")
arr2 = np.array([list(map(int, input().split()))
```

Debugger

Average time	Maximum time
0.051 s	0.065 s
50.75 ms	65.00 ms

2 out of 2 shown test case(s) p

2 out of 2 hidden test case(s) p

Test case 1 65 ms Debug

Expected output	Actual output
Enter Array1:	Enter Array1:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Array2:	Enter Array2:
4 5 6	4 5 6

Terminal Test cases

< Prev Reset Submit Next >

Essentials of Data Science Laboratory - 2304102L - 2026
Search course
3. Practical 3
3.1. Practice Lab Assignment
3.1.1. Numpy Array Operations
3.2. Lab Assignment
3.2.1. Numpy: Matrix Operations
3.2.2. Numpy: Horizontal and Vertical Stac...
3.2.3. Numpy: Custom Sequence Generation
3.2.4. Numpy: Arithmetic and Statistical Op...
3.2.5. Numpy: Copying and Viewing Arrays
3.2.6. Numpy: Searching, Sorting, Countin...
3.2.7. Student Data Analysis and Operations
4. Practical 4
5. Practical 5

3.2.3. Numpy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using `numpy` based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Sample Test Cases +

customS...

```python
import numpy as np

# Take user input for the start, stop, and step of the sequence
start = int(input())
stop = int(input())
step = int(input())

# Generate the sequence using np.arange()
```

Average time	Maximum time
0.037 s (37.00 ms)	0.052 s (52.00 ms)

2 out of 2 shown test case(s) p
2 out of 2 hidden test case(s) p

Test case 1 28 ms Debug

Expected output	Actual output
3	3
10	10
2	2
[3 5 7 9]	[3 5 7 9]

Test case 2 52 ms

Terminal Test cases

< Prev Reset Submit Next >

Essentials of Data Science Laboratory - 2304102L - 2026

Pin

Search course ctrl + k

3. Practical 3

3.1. Practice Lab Assignment

3.1.1. Numpy Array Operations

3.2. Lab Assignment

3.2.1. Numpy: Matrix Operations

3.2.2. Numpy: Horizontal and Vertical Stac...

3.2.3. Numpy: Custom Sequence Generation

3.2.4. Numpy: Arithmetic and Statistical Op...

3.2.5. Numpy: Copying and Viewing Arrays

3.2.6. Numpy: Searching, Sorting, Countin...

3.2.7. Student Data Analysis and Operations

4. Practical 4

5. Practical 5

3.2.4. Numpy: Arithmetic and Statistical Operations, M... 26:27

You are given two arrays `A` and `B`. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:
- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:
- Calculate the mean, median, and standard deviation of array `A`.

3. Bitwise Operations:
- Perform bitwise `AND`, bitwise `OR`, and bitwise `XOR` on the arrays (ex: A_i OR B_i).

Input Format:
- The first line contains space-separated integers representing the elements of array `A`.
- The second line contains space-separated integers representing the elements of array `B`.

Output Format:
- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Explorer | different... | Submit | Debugger

```python
import numpy as np

def array_operations(A, B):

    # Convert A and B to NumPy arrays
    A= np.array(A)
    B= np.array(B)
    # Arithmetic Operations
    sum_result = A + B
```

Average time	Maximum time
0.020 s	0.030 s
19.67 ms	30.00 ms

1 out of 1 shown test case(s) p

2 out of 2 hidden test case(s) p

Test case 1 30 ms Debug

Expected output	Actual output
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
Element-wise Sum: 6 8 10 12	Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4	Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32	Element-wise Product: 5 12 21 32

Terminal | Test cases

< Prev | Reset | Submit | Next >

Essentials of Data Science Laboratory - 2304102L - 2026 Pin
Search course ctrl + k
3. Practical 3
3.1. Practice Lab Assignment
3.1.1. Numpy Array Operations
3.2. Lab Assignment
3.2.1. Numpy: Matrix Operations
3.2.2. Numpy: Horizontal and Vertical Stac...
3.2.3. Numpy: Custom Sequence Generation
3.2.4. Numpy: Arithmetic and Statistical Op...
3.2.5. Numpy: Copying and Viewing Arrays
3.2.6. Numpy: Searching, Sorting, Countin...
3.2.7. Student Data Analysis and Operations
4. Practical 4
5. Practical 5

3.2.5. Numpy: Copying and Viewing Arrays

01:35

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

```
Original array after modifying view: <original_array>
View array: <view_array>
```

- After modifying the copy:

```
Original array after modifying copy: <original_array>
Copy array: <copy_array>
```

copyAnd... Submit

```python
import numpy as np

inputlist = list(map(int,input().split(" ")))

# Original array
original_array = np.array(inputlist)

# Create a view
view_array = original_array.view()
```

Average time 0.012 s 12.25 ms
Maximum time 0.017 s 17.00 ms

2 out of 2 shown test case(s) p
2 out of 2 hidden test case(s) p

Test case 1 17 ms Debug

Expected output	Actual output
10 20 30 40 50 60 70 80	10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]	Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]	View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]	Original array after modifying copy: [99 20 30 40 50 60 70 80]

Terminal Test cases

< Prev Reset Submit Next >

Essentials of Data Science Laboratory - 2304102L - 2026
Pin
Search course ctrl + k
3. Practical 3
3.1. Practice Lab Assignment
3.1.1. Numpy Array Operations
3.2. Lab Assignment
3.2.1. Numpy: Matrix Operations
3.2.2. Numpy: Horizontal and Vertical Stac...
3.2.3. Numpy: Custom Sequence Generation
3.2.4. Numpy: Arithmetic and Statistical Op...
3.2.5. Numpy: Copying and Viewing Arrays
3.2.6. Numpy: Searching, Sorting, Countin...
3.2.7. Student Data Analysis and Operations
4. Practical 4
5. Practical 5

3.2.6. Numpy: Searching, Sorting, Counting, Broadcas... 01:24

The given code in the editor takes a single array, `array1`, as space-separated integers as input from the user.
Additionally, it takes the following inputs:

- `search_value`: The value to search for in the array.
- `count_value`: The value to count its occurrences in the array.
- `broadcast_value`: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:
1. Searching: Find the indices where `search_value` appears in `array1` and print these indices.
2. Counting: Count how many times `count_value` appears in `array1` and print the count.
3. Broadcasting: Add `broadcast_value` to each element of `array1` using broadcasting, and print the resulting array.
4. Sorting: Sort `array1` in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing `array1`.
2. An integer `search_value` represents the value to search for in the array.
3. An integer `count_value` represents the value to count in the array.

Explorer arrayOpe... Submit Debugger

```
import numpy as np

# Input array from the user
array1 = np.array(list(map(int, input().split())))

# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))
```

Average time 0.039 s 39.25 ms
Maximum time 0.045 s 45.00 ms
2 out of 2 shown test case(s) p
2 out of 2 hidden test case(s) p

Test case 1 45 ms Debug

Expected output	Actual output
1 1 1 2 2 2	1 1 1 2 2 2
Value to search: 1	Value to search: 1
Value to count: 2	Value to count: 2
Value to add: 2	Value to add: 2
[0 1 2]	[0 1 2]
3	3

Terminal Test cases

< Prev Reset Submit Next >

CODETANTRA Home Learn Anywhere 202501040161@mitaoe.ac.in Support Logout

Essentials of Data Science Laboratory - 2304102L - 2026 Pin

Search course ctrl + k

- 3. Practical 3
  - 3.1. Practice Lab Assignment
    - 3.1.1. Numpy Array Operations
  - 3.2. Lab Assignment
    - 3.2.1. Numpy: Matrix Operations
    - 3.2.2. Numpy: Horizontal and Vertical Stac...
    - 3.2.3. Numpy: Custom Sequence Generation
    - 3.2.4. Numpy: Arithmetic and Statistical Op...
    - 3.2.5. Numpy: Copying and Viewing Arrays
    - 3.2.6. Numpy: Searching, Sorting, Countin...
    - 3.2.7. Student Data Analysis and Operations
- 4. Practical 4
- 5. Practical 5

3.2.7. Student Data Analysis and Operations 01:06

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details**: Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students**: Determine the total number of students in the dataset.
- **Print all student roll numbers**: Extract and print the roll numbers of all students.
- **Print Subject 1 marks**: Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2**: Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3**: Identify the highest marks in Subject 3.
- **Print all subject marks**: Display the marks of all students for each subject.
- **Find total marks of students**: Compute the total marks for each student across all subjects.
- **Find the average marks of each student**: Compute the average marks for each student.
- **Find average marks of each subject**: Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2**: Compute the

Explorer | Operatio... | Submit | Debugger

```python
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',',
skiprows=1)

# 1. Print all student details
print("All student Details:\n", a )

# 2. print total students
```

Average time 0.049 s 49.00 ms | Maximum time 0.049 s 49.00 ms | 1 out of 1 shown test case(s) p

Test case 1 49 ms Debug

Expected output	Actual output
All student Details:	All student Details:
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[303. 45. 56. 89.]	[303. 45. 56. 89.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]

Terminal | Test cases

< Prev | Reset | Submit | Next >